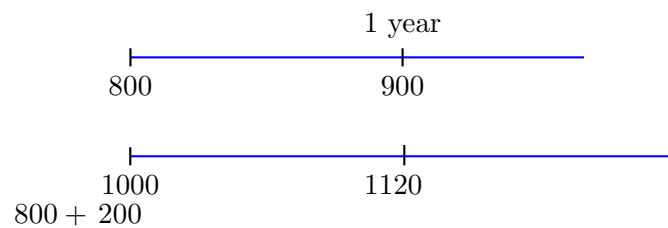


Problem 3.9

John wants to have \$800. He may obtain it by promising to pay \$900 at the end of one year; or he may borrow \$1,000 and repay \$1,120 at the end of the year. If he can invest any balance over \$800 at 10% for the year, which should he choose?

Solution

We compare the two options.



For the first option,

$$900 = 800(1 + i).$$

So,

$$i = \frac{900}{800} - 1 = 0.125 = 12.5\%.$$

Now consider the second option. If John borrows \$1000, he uses \$800 and invests the remaining \$200 at 10%. Thus, after one year the investment becomes

$$200(1 + 0.10) = 220.$$

So he earns \$20 from the investment.

He must repay \$1120. Since

$$1120 - 1000 = 120,$$

and he earned \$20 from investing the extra \$200, the net cost is

$$120 - 20 = 100.$$

Therefore, the effective rate relative to the \$800 he needed is

$$i = \frac{100}{800} = 0.125 = 12.5\%.$$

Hence, it does not matter which option he chooses; both have the same net effective interest rate.